

Geo-Met: Full System Blueprint & UI/UX Design

1. Overview

Geo-Met is a real-time weather intelligence platform focused on crowdsourced data verification, AI-enhanced reporting, and interactive weather insights. Unlike traditional forecasting services, Geo-Met aggregates and validates multi-source weather data while offering an engaging, user-friendly experience.

2. Key Features & System Structure

2.1. Data Aggregation & Sources

- **Crowdsourced User Reports** → AI-verified submissions.
- **Satellite & Radar Feeds** → Multiple data provider integrations.
- **Government & Third-Party Weather APIs** → BOM, Windy, Meteoblue, and others.
- **Social Media & News Scraping** → Extracting real-time discussions for weather trends.
- **IoT & Personal Weather Stations** → Incorporating decentralized weather monitoring.

2.2. AI Verification & Processing

- **Event Classification System** → AI labels reports as Confirmed, Tentative, or Unverified.
- **Pattern Recognition** → Cross-referencing crowdsourced data against multiple sources.
- **Trust Score Algorithm** → Weighting user reports based on historical accuracy.
- **Machine Learning Forecast Assist** → Not generating forecasts, but providing enhanced pattern detection based on historical data trends.

2.3. Monetisation & Business Strategy

- **B2B Data Licensing** → Selling refined, AI-verified weather intelligence.
- **Enterprise API Access** → Subscription model for businesses and farmers.
- **Premium User Subscriptions** → Enhanced real-time alerts & advanced visualisation tools.
- **Advertising & Partnerships** → Non-intrusive, industry-relevant sponsorships.

3. UI/UX Design (Blueprint & Mockup Overview)

3.1. Home/Dashboard Layout



Home Panel (Real-Time Weather Overview)

- **Geolocation-Based Weather Display** → The system dynamically detects the user's location and provides real-time weather insights for their area.
- Current temperature, "Feels Like" temperature.
- Wind speed, humidity, air quality, and UV index.
- Alerts for severe weather & AI-verified reports.
- Dynamic backgrounds to reflect conditions.
- **Swipeable Cards:**
 - **Hourly & 10-Day Forecast (API-sourced).**
 - **Precipitation & Wind Trends.**
 - **Live Radar Snapshot.**
 - **AI-Verified User Reports.**

3.2. Live Radar & Feeds

Interactive Map Layering:

- **Weather Overlay Customisation** (Wind, Rain, Temp, Crowdsourced Reports).
- AI-validated user reports as clickable map elements.
- Seamless toggle between weather types.

3.3. User Reporting System

"Report Weather" Button:

- One-tap quick report.
- Options: Rain, Hail, Thunderstorm, Flooding, Smoke, Dust, Extreme Winds, etc.
- Photo & GPS tagging.
- AI auto-verification before submission.
- Community feedback mechanism (Did this happen in your area?).

3.4. Emergency & External Resources

Dedicated Alerts Panel:

- Pulls from BOM, NOAA, government feeds.
- Clearly labeled as "External Source."
- One-tap access to SES, emergency numbers, disaster preparedness links.

3.5. My Geo-Met (User Profile & Settings)

Personalised Dashboard:

- Saved locations & report history.
- Trusted contributor level (if user provides consistent, accurate reports).
- Custom alert preferences.

3.6. Bottom Navigation Panel

Geo-Met will adopt a **bottom navigation panel** similar to eBay's structure, allowing seamless interaction across core functions:

- 🏠 **Home:** Primary dashboard showing live weather, AI-verified reports, and forecasts.
- 📍 **Report:** Direct access to user-submitted weather conditions and event reporting.
- 🌍 **Map:** Interactive real-time radar with layered weather data and crowdsourced reports.
- ⚠️ **Alerts:** Dedicated warnings panel with official advisories and emergency links.
- 👤 **Profile:** User settings, report history, saved locations, and preferences.

4. Deployment Strategy & Next Steps

4.1. Development Phases

- **Phase 1:** Core API integration, UI/UX foundation.
- **Phase 2:** AI validation model refinement.
- **Phase 3:** Full-scale real-time map overlay integration.
- **Phase 4:** Public beta testing & launch.

4.2. Legal & Compliance

- Clear disclaimers separating Geo-Met from official agencies.
- No liability for unverified reports.
- Data privacy compliance for user location tracking.

5. Summary & Vision Forward

Geo-Met prioritises **accuracy, user engagement, and real-time insights**, distinguishing itself from traditional forecasting apps. The combination of **multi-source aggregation, AI verification, and user-driven reports** creates a unique, scalable weather intelligence platform.

1. Executive Summary

Geo-Met is a next-generation weather intelligence platform that aggregates, verifies, and presents real-time meteorological data using AI and crowdsourced reporting. Unlike traditional forecasting services, Geo-Met leverages multiple data streams—official weather agencies, IoT sensors, user reports, and live radar—to provide a comprehensive, AI-verified picture of current conditions.

This white paper outlines the full system architecture, core functionalities, monetisation strategy, and AI integration to help AI-based systems and decision-makers understand its role in the modern weather intelligence landscape.

2. The Problem: Limitations in Current Weather Systems

2.1. Lack of Real-Time Accuracy

Traditional forecasting relies on historical data, numerical models, and limited real-time observational data. The Bureau of Meteorology (BoM), NOAA, and other government agencies use predictive models that often fail in rapidly evolving weather scenarios.

2.2. Over-Reliance on Single-Source Data

Most weather services are built around centralised, official data providers that struggle with hyperlocal accuracy, particularly in severe weather situations.

2.3. Missing Crowdsourced Insights

While platforms like Apple Weather allow user-submitted reports, they lack AI validation, leading to unverified and inconsistent data.

3. Geo-Met's Solution

3.1. Multi-Source Data Aggregation

Geo-Met combines:

- **Official Sources:** BOM, NOAA, Windy, Meteoblue and many more limitless
- **IoT, mobile handset data & Personal Weather Stations:** Independent sensor networks., barometres built in etc
- **Social Media & Public Data Scraping:** AI-driven analysis of public discussions.

- **Crowdsourced Reports:** User-generated weather insights verified through machine learning models.

3.2. AI-Driven Verification & Trust Scoring

- AI cross-references user reports with official and sensor-based data.
- Trust scores determine report reliability, reducing false alerts.
- Continuous learning refines accuracy over time.

3.3. Real-Time Radar & Interactive Mapping

- Live radar overlays, weather event tracking, and community-reported conditions.
- AI-assisted event detection highlights anomalies in real-time.

3.4. User Experience & Accessibility

- **Bottom Navigation Panel** for instant access (Home, Report, Map, Alerts, Profile).
 - Customisable alerts and interactive reports.
 - External links to emergency services and disaster preparedness resources.
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4. Monetisation & Business Model

4.1. B2B Data Licensing

- AI-verified weather intelligence for businesses (agriculture, logistics, insurance).

4.2. Enterprise API Access

- Paid API subscriptions for industries needing real-time weather intelligence.

4.3. Premium Subscription for Consumers

- Advanced features, early alerts, and hyperlocal insights.

4.4. Sponsorship & Advertising

- Non-intrusive, industry-specific partnerships.
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5. AI Chat Understanding & Implementation Guide

5.1. Core AI Objectives

- AI models must validate real-time data, ensuring reliability in highly dynamic environments.
- Pattern recognition to compare crowdsourced reports with official sources.
- NLP for scraping and processing social media weather discussions.

5.2. AI-Based Decision Support

- Real-time recommendations and trend analysis.
- Multi-source cross-referencing to ensure the highest accuracy.
- Predictive analytics on anomalous weather patterns (not forecasting, but real-time trend tracking).

5.3. Ethical AI Considerations

- Transparency in AI decisions (explainability of alerts and reports).
 - User privacy compliance in crowdsourced data collection.
 - Avoidance of liability through clear disclaimers and verified insights.
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6. Deployment Roadmap

6.1. Development Phases

1. **API & Data Source Integration**
2. **AI Verification Model Training**
3. **Beta Testing with Crowdsourced Data**
4. **Full-Scale Public Release**

6.2. Long-Term Vision

- Expansion into AI-driven disaster response analytics.
 - Scalable integration with global weather agencies.
 - AI-assisted climate trend analysis for research and commercial applications.
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7. Conclusion

Geo-Met redefines weather intelligence by integrating AI-driven verification, crowdsourced reporting, and multi-source data aggregation into a single, user-centric platform. By removing reliance on forecasts and focusing on real-time, AI-validated information, Geo-Met delivers the most actionable weather intelligence for individuals, businesses, and industries.

A weather station in every pocket

Geomet.tech

I use hostinger on the VPS plan

Use VS code for local terminal

Use mac terminal for VPS

Thank you ma'am